

1. We expect the human capital indicators to move with food price inflation. Recent evidence confirms that human capital is a core productive capacity behind agricultural productivity growth in developing economies (e.g. Frontiers in Sustainable Food Systems, 2025)¹. Productivity is the price mechanism: Fuglie and Wang (2012)² show that sustained productivity growth is the main reason real food prices trended downward globally through the twentieth century. Higher human capital stock, therefore, should be associated with lower and more stable food inflation over the medium run.
2. Ideally, we would use the Human Capital Index Plus indicator from the World Bank, along with its dimensions such as education, health and employment. However, the data is mostly missing as this is a newly established indicator. Therefore, we will use proxies for education and health that we collected from the World Bank: life expectancy and tertiary school enrollment.
3. Regression analysis:
 - Education has a negative relationship with food price inflation (coef -0.2575). However, the relationship is not significant as the p-value of 0.356 is greater than 0.05.
 - Life expectancy has a small positive relationship with food inflation (coef 0.0618). This could be because as people live longer, there is greater demand for food, which drives food prices up. The relationship is not significant as the p-value of 0.850 is greater than 0.05.
 - The r-squared of 0.087 is very low, which is expected of social science data.
 - The Durbin-Watson statistic of 0.946 indicates positive autocorrelation in the regression residuals. Therefore, the reported standard errors and significance tests should be interpreted with caution. Given the limited annual sample (22 observations) and the exploratory objective of quantifying the relationship between food inflation and human capital indicators, more advanced corrections for serial correlation were not implemented.

OLS Regression Results						
Dep. Variable:	BWA_FAO_CP_23014	R-squared:	0.087			
Model:	OLS	Adj. R-squared:	-0.009			
Method:	Least Squares	F-statistic:	0.9099			
Date:	Fri, 24 Jul 2026	Prob (F-statistic):	0.419			
Time:	22:02:44	Log-Likelihood:	-64.216			
No. Observations:	22	AIC:	134.4			
Df Residuals:	19	BIC:	137.7			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	7.9974	15.912	0.503	0.621	-25.307	41.302
BWA_education	-0.2575	0.272	-0.946	0.356	-0.827	0.312
BWA_life_expectancy	0.0618	0.323	0.191	0.850	-0.615	0.739
Omnibus:	6.359	Durbin-Watson:	0.946			
Prob(Omnibus):	0.042	Jarque-Bera (JB):	4.232			
Skew:	1.014	Prob(JB):	0.121			
Kurtosis:	3.712	Cond. No.	977.			

Notes:

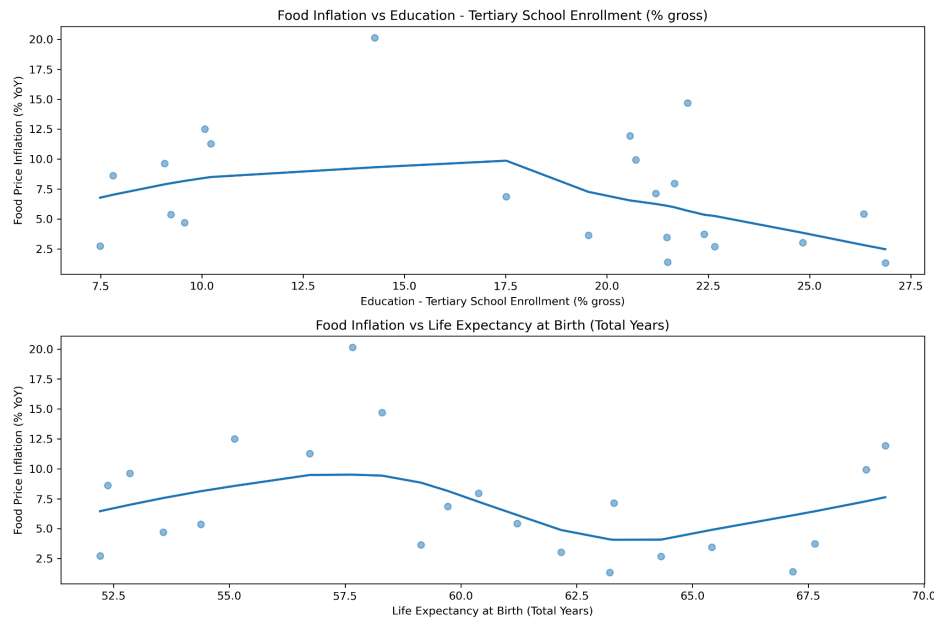
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

- Lowess plots suggest a polynomial relationship between food price inflation and life expectancy. While the lowess plots do not explicitly acknowledge a temporal dimension, they account for time

¹ Role of productive capacities on agricultural productivity in BRI countries (2025). Frontiers in Sustainable Food Systems.

² Fuglie, K. & Wang, S.L. (2012). Productivity Growth in Global Agriculture Shifting to Developing Countries. Choices, Quarter 4.

series data by treating the time index as a standard independent variable, allowing a non-parametric method to reveal underlying trends.



4. Granger Causality Test

- The Granger causality test indicates that education has predictive power for food inflation at a one-year lag ($F = 6.32$, $p = 0.022$, 5% significance). Therefore, the null hypothesis that education does not Granger-cause food inflation is rejected. In contrast, life expectancy does not significantly improve predictions of food inflation at any lag (lag 1: $F = 0.53$, $p = 0.48$).

5. Forward projection

- The forward projection shows sharp decline from 2024, followed by a slow decline of food price inflation. The levels are also similar to those of the food price inflation predicted with Baltic Dry Index, Brent crude oil prices and policy rate, which should give us a little confidence in our results. Further, the forward projections for education and life expectancy show a steady increase over 5 years, confirming our initial background research that higher human capital levels should be associated with lower food prices. Refer to the visualizations submission for the full forecasts.

